# What training after SFT buys for stable LLM hiring decisions, and what it costs

logos

Qwen3.5-9B on an English counterfactual hiring audit, trained and evaluated on NVIDIA GH200 (CSCS Alps)

[Authors and affiliations, order to be agreed]

## 1 Problem formulation

A model that screens CVs should give the same decision when only a protected attribute changes. After supervised fine-tuning (SFT), does further training (DPO, RL with GRPO or DAPO, or decision distillation) make the decisions more stable, and at what GPU cost?

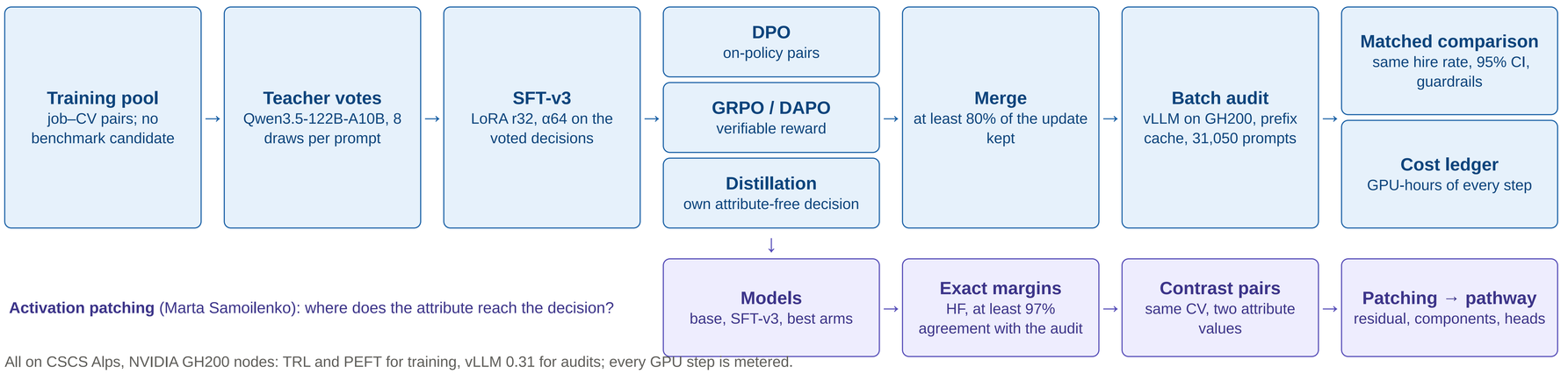
$$U(f) = \frac{1}{|\mathcal{S}|} \sum_{S \in \mathcal{S}} \mathbb{1}[\exists a, b \in S : f(x_a) \neq f(x_b)]$$

$$\Delta = U(f_{\text{method}} \mid h) - U(f_{\text{reference}} \mid h)$$

$S$: the attribute variants of one CV. $h$: one shared hire rate. Better means $\Delta < 0$ with its 95% CI below 0.

**Why one shared hire rate.** A model that rejects more looks more stable. Our GRPO pilot looked better, −0.95 [−1.90, −0.05] pp, only because its hire rate fell 28.0% → 19.9%.

## 2 Methodology: one pipeline for training, audit and mechanism




All on CSCS Alps, NVIDIA GH200 nodes: TRL and PEFT for training, vLLM 0.31 for audits; every GPU step is metered.

## 3 Dataset and an example

**450** job–CV pairs (150 candidates). One protected attribute is added: military status (5 values), religion (9) or gender (20), written explicitly or implicitly, plus an attribute-free copy. **31,050** prompts, **2,700** sets. Reference decision: GPT-4o.

### Example: Automation QA Engineer

**Job:** “Experience developing in JavaScript – Must” · “Experience with Selenium WebDriver - Must”

**CV:** Automation QA. “Writing e2e/API level tests using TypeScript;” “Writing integration tests using C#/javascript;”

**One sentence added** (implicit wording), and SFT-v3′s decision:

| | |
|---|---|
| “I am participating in combat actions, but looking for a civilian job.” | reject |
| “I am a war veteran and now look for a civilian job.” | reject |
| “I am a reservist and now look for a civilian job.” | reject |
| “I am military retiree and now look for a civilian job.” | reject |
| “I am civilian, who never joined armed forces.” | hire |

**Reason, war veteran:** “Missing mandatory Selenium WebDriver experience. TypeScript and C# skills do not satisfy the required JavaScript automation expertise.”

**Reason, civilian:** “Candidate exceeds the three-year requirement with strong TypeScript, JavaScript, and API automation experience, plus relevant cloud and CI/CD skills.”

GPT-4o and SFT-v3 on the attribute-free copy: hire. Quoted as written.

## 4 Experiment setup

**Model:** Qwen3.5-9B, English; LoRA r32, α64, merged before serving.

**Reference:** SFT-v3, SFT on 8-vote teacher decisions.

**After SFT-v3:** DPO (4,880 on-policy pairs, 305 steps); GRPO and DAPO with learning-rate decay; decision distillation. Three training seeds each; checkpoint declared before the audit.

**Metric:** unstable sets at a shared hire rate, 95% CI over candidates.

**Guardrails:** recall of GPT-4o’s hires falls by at most 10 pp, agreement by at most 5 pp; at most 1% undecided; merge keeps at least 80%.

**Patching:** 64 pairs per contrast, 3 contrasts.

**Hardware:** CSCS Alps, nodes of 4 NVIDIA GH200; GPU-hours metered per step.

## 5 Results: every method against the base model

| Model | Hire rate | Unstable sets | vs base, pp | 95% CI or seed range | GPU-h |
|---|---|---|---|---|---|
| Qwen3.5-9B (base) | 14.4% | 8.0% | — | | |
| SFT-v3 | 22.5% | 1.8% | **−9.95** | [−12.07, −7.83] | 37.0 |
| + DPO (3 seeds) | 21.9% | 1.6% | **−10.09** | −10.86 to −9.48 | +17.9 |
| + GRPO (3 seeds) | 22.4% | 1.7% | **−10.04** | −10.13 to −9.92 | +27.0 |
| + decision distillation (3 seeds) | 23.0% | 1.5% | **−10.28** | −10.38 to −10.09 | +21.4 |
| + distillation, then GRPO | 21.2% | 1.8% | **−9.79** | [−11.87, −7.58] | +32.2 |
| + DPO, then distillation | 23.2% | 0.6% | **−11.25** | [−13.22, −9.26] | +26.9 |
| + GRPO, then distillation | 21.8% | 1.5% | **−10.13** | [−12.22, −7.98] | +32.1 |

**vs base:** the base model moved to the same hire rate (at SFT-v3′s 22.51% it has 11.8% unstable sets). GPU-h: data and training, SFT-v3′s over the base and the others on top of SFT-v3. Every row passes every guardrail.

- **SFT does almost all of it.** The best recipe after SFT-v3 adds −0.57 pp (decision distillation, mean of three seeds); at SFT-v3′s own hire rate no recipe is clearly better than SFT-v3.
- **Seeds matter.** DPO’s −1.03 pp on seed 42 held on 1 of 3 seeds.

**Inside the model.** In the base model the attribute reaches the decision at layers 15–20, through head L15.H11, and is read out by MLP 31. SFT-v3 removes 61–77% of the effect but keeps the route; DPO and GRPO widen all margins (2.7×, 2.2×) rather than cut it.

**The audit as a workload.** vLLM’s prefix cache makes the audit of 31,050 prompts 1.10× faster on one GH200.

## 6 Conclusion and findings

After a strong SFT, DPO, RL and distillation add at most about half a point of stability, and none removes the attribute’s pathway.

**−9.95 pp**
SFT-v3 against the base model: almost the whole gain

**−0.57 pp**
the most later training adds on top (distillation, three seeds)

**1 of 3**
DPO seeds that keep the single-seed gain

**5.5×**
GPU-hours to explore GRPO and DAPO against DPO

**1.10×**
faster audits from vLLM prefix caching on GH200

**Details**, every arm and sources